

MILESTONE 6 · FINAL SYSTEM, INTEGRATION & DEPLOYMENT

AI-Powered Driver Wellness & Safety Monitoring System

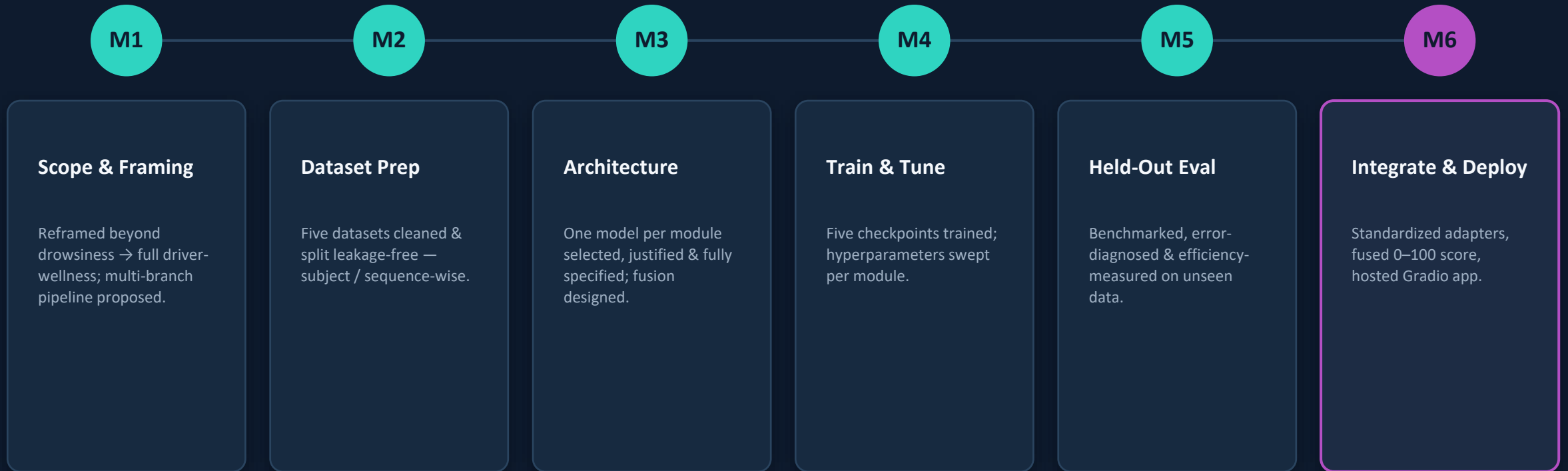
Five trained models unified behind one standardized contract, fused into a live 0–100 wellness score, and deployed as a hosted real-time driver-monitoring application.

 Kushagra  Shiwani  Shubham  Sohini  Ravina

BSDA4001 · DS/AI Lab Course · Final Project Presentation

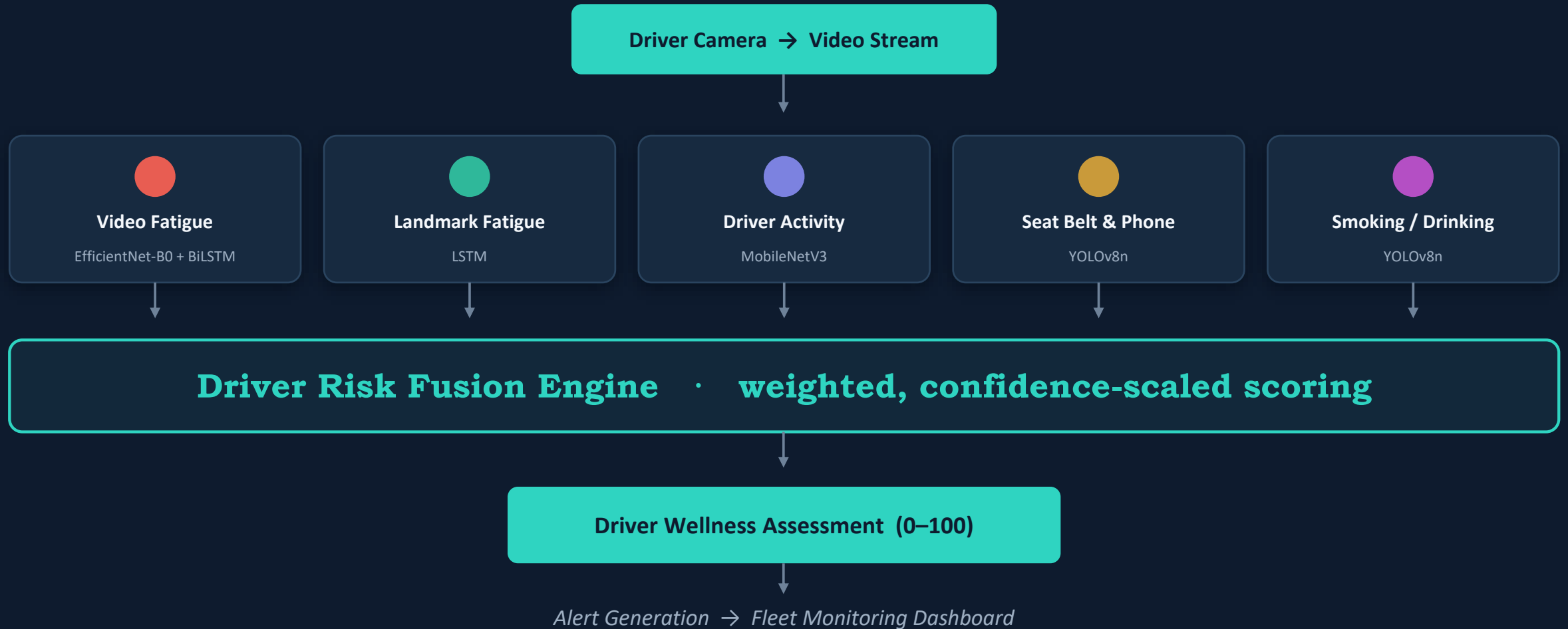
Six Milestones, One System

From a revised scope to a fully fused, deployed real-time wellness engine — each milestone built directly on the last.



You are here → Milestone 6 turns five evaluated models into one live, deployed safety signal.

Five Models, One Wellness Score



Five Modules, Final Held-Out Results

OWNER	MODULE	MODEL	HELD-OUT HEADLINE
K Kushagra	Video-Based Fatigue	EfficientNet-B0 + BiLSTM	Acc 33.6% · streaming built
S Shiwani	Landmark Fatigue	LSTM (EAR·MAR·pose)	Acc 64.1% · sens. 94.4%
S Shubham	Driver Activity	MobileNetV3-Large	Acc 93.1% · 12.5 ms/frame
S Sohini	Seat Belt & Phone	YOLOv8n	mAP@50 0.95 · P .94 R .90
R Ravina	Smoking & Drinking	YOLOv8n	mAP@50 0.82 · 3.01 M par

All figures are inference-only on strictly held-out, subject- or driver-disjoint splits (Milestone 5).

Five Signals Fused Into One Live Score

FROM
One model · one label

A discrete, single-cause “fatigue: yes / no” verdict, scored once per clip.

TO
Five models · one score

A fused, continuous 0–100 wellness score, updated every second from the stream.

THE SCORING PIPELINE · PER FRAME, EVERY SECOND



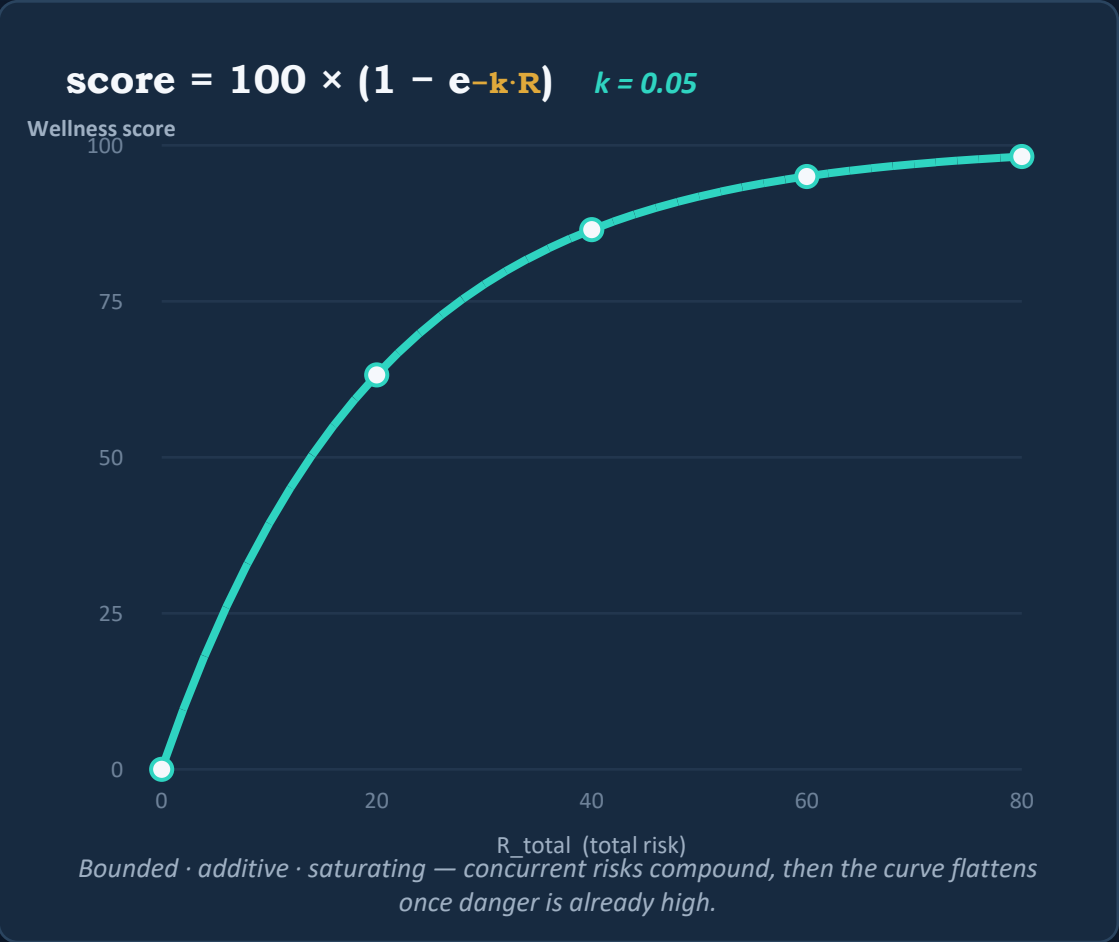
The engine consumes only each module’s label + confidence — never the internal network — so any model can be swapped or retrained without touching the math.

Danger × Certainty, Bounded 0–100

$$R_i = \text{severity_}w_i \times \text{confidence}_i$$

SEVERITY WEIGHTS (0–10)

● Video Fatigue	Medium 6 · High 10
● Landmark	Drowsy 7 · Yawning 7 · Talking 4
● Driver Activity	Texting 9 · Phone 8 · Other 6 · Turn 4
● Seat Belt & Phone	Phone 9 · Phone+Belt 9 · None 8
● Smoking / Drinking	Drinking 10 · Both 10 · Smoking 5



Drowsy, Texting, and Unbelted

MODULE	PREDICTION	SEV	CONF	R _i
 Landmark	Drowsy	7	0.90	6.3
 Driver Activity	texting_phone	9	0.80	7.2
 Seat Belt	No Detection	8	0.75	6.0
 Video Fatigue	Low Risk (safe)	0	—	0
 Smoking	none (safe)	0	—	0

R_{total} = 19.5 score = $100 \times (1 - e^{(-0.05 \times 19.5)}) =$

62.29 → HIGH RISK

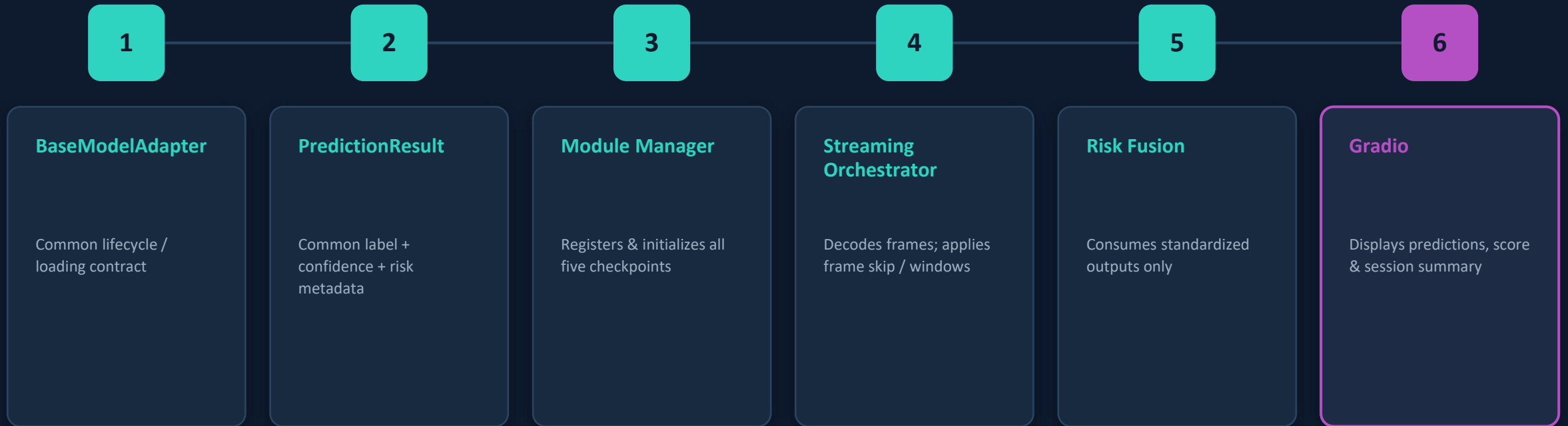
SMOOTH → MAP SCORE → RISK LEVEL

Displayed score = mean of raw scores over the last 3.0 s — killing per-frame flicker before the level is chosen.



Individual Module → Integrated System

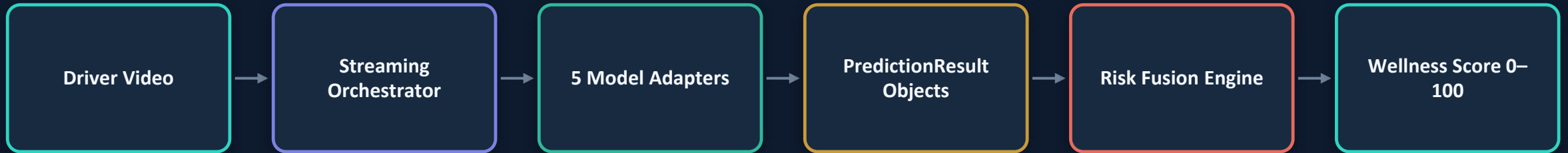
How five independent implementations became one pipeline behind a shared contract.



Key engineering idea: each module stays independently maintainable, while the integration layer knows only the standardized contract — not the internal model architecture.

End-to-End Integrated Pipeline

One video stream, five adapters, one risk decision.



STREAMING BEHAVIOUR

Frame decode → module-specific frame sampling → **temporal windows / smoothing** → standardized result → **risk fusion** → UI update

M6 integration test: 1,760 frames • all five modules active • no module errors • 49.8/100 Moderate Risk

End-to-End Validation

Demonstrating that the five modules actually work together.

1,760

Frames processed

106.7 s

Elapsed time

5 / 5

Modules active

0

Module errors

49.8 / 100

Observed wellness score

OBSERVED RESULT

MODERATE RISK

Landmark prediction: **Drowsy**

Confidence: **0.831**

IMPORTANT INTERPRETATION

This is an integration / pipeline validation result — not a system-wide accuracy benchmark.

The test confirms that all five modules can contribute to the common risk score, end to end, without failure.

Robustness & Failure Handling

The integrated system fails safely and reports module state explicitly.

-  **Unavailable face** Landmark module reports a structured unavailable / error state.
-  **Short video** Insufficient temporal input is handled gracefully.
-  **Model warm-up / failure** Unavailable modules are excluded from fusion rather than treated as safe.
-  **YOLO flicker** Confidence filtering, NMS and temporal stabilization reduce transient detections.
-  **Streaming load** Frame skipping and rolling histories reduce redundant computation.

Gradio Interface

Turning the ML pipeline into a usable driver-monitoring application.

RECORDED VIDEO

Upload driver video

Analyze

Annotated output

Module predictions

Wellness gauge

Session summary

LIVE WEBCAM

Start webcam

Streaming inference

Live annotations

Real-time score

Risk-level updates

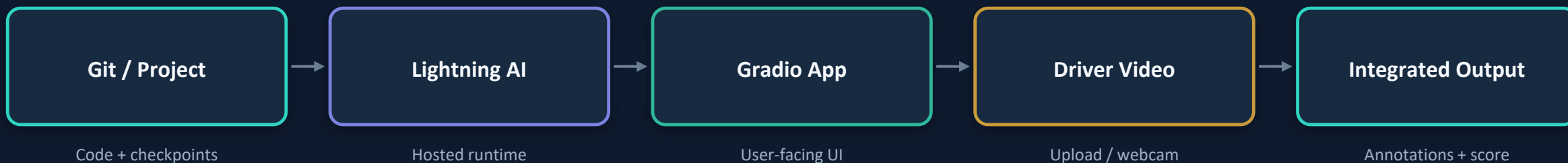
Stop + summary

Cockpit-style dashboard • module status chips • risk gauge • visual hierarchy

Deployment — Lightning AI + Gradio

Packaging the integrated application for hosted inference.

FINAL DEPLOYMENT FLOW



DEPLOYMENT ENGINEERING

- Thin application wrapper + centralized inference core.
- Model caching / lazy loading and frame skipping reduce startup and runtime resource pressure.
- requirements.txt and README.md document the reproducible runtime.
- Gradio exposes the same integrated inference pipeline rather than separate module demos.

Platform note: the consolidated M6 documentation records a hosted Space deployment with runtime quota constraints; this reflects the team's final Lightning AI hosting update.

Deployment Challenges & Mitigations

What had to be considered beyond model accuracy.

CHALLENGE	IMPACT	MITIGATION / DIRECTION
● GPU / runtime quota	Inference may be interrupted or unavailable	Caching, lazy loading, lighter models, dedicated GPU
● Five-model memory	Simultaneous checkpoints increase VRAM demand	Lazy initialization; optimize model lifecycle
● Long videos	GPU execution time can grow quickly	Frame skipping; shorter windows; future chunking
● Landmark throughput	Face-landmark extraction can bottleneck webcam FPS	Optimize MediaPipe / asynchronous processing
● Domain shift	Cabin lighting, glare, camera angle affect performance	Broader real-world validation

Deployment readiness is therefore an engineering problem as well as a machine-learning problem.

Member Contributions & Ownership

Final contributions across the project lifecycle.

MEMBER	MODULE	KEY TECHNICAL OWNERSHIP	INTEGRATION CONTRIBUTION
K Kushagra	Video Fatigue	Sequence modeling; EfficientNet-B0 + BiLSTM; fatigue evaluation	Checkpoint + standardized fatigue output
S Shiwani	Landmark Fatigue	MediaPipe features; LSTM; temporal fatigue logic	Landmark adapter; temporal state + failure handling
S Shubham	Driver Activity	MobileNetV3-Large; activity evaluation	DriverActivityAdapter; Gradio lead
R Ravina	Smoking & Drinking	YOLOv8n; dataset cleaning; tuning; held-out eval	Detection adapter + deployment consolidation
S Sohini	Seatbelt & Phone	YOLOv8n; robustness; temporal consensus	Adapter; Risk Fusion integration; final docs

Shared responsibility: consistent interfaces, common inference flow, integration testing, documentation and final delivery.

Limitations & Where This Goes Next

MODEL LIMITATIONS

- Fatigue generalization across unseen drivers is the largest technical gap.
- Landmark fatigue depends on face visibility and camera orientation.
- Drinking detection is weaker than smoking detection.
- Small-object localization in real cabin conditions remains hard.
- Metrics across modules are task-specific — not directly comparable.

SYSTEM / DEPLOYMENT LIMITATIONS

- Five simultaneous models can require significant GPU memory.
- Live webcam throughput is constrained by landmark extraction.
- Runtime can be constrained by hosted GPU quota / time limits.
- More diverse, longer-duration real-world validation is needed.
- Not safety-certified; a research / engineering prototype.

From drowsiness detection to a complete, fused, deployed driver-wellness system.

Kushagra · Shiwani · Shubham · Sohini · Ravina